

KON 313

Feedback Control Systems

Modeling of the Mechanical Systems in Frequency Domain

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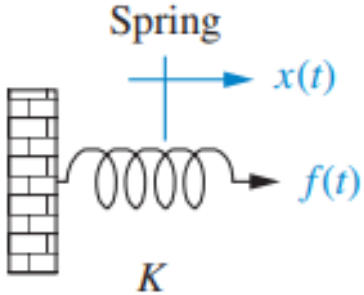
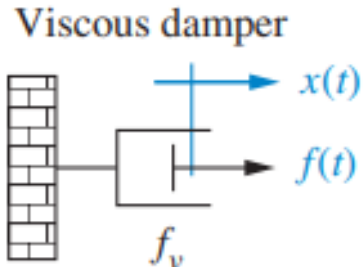
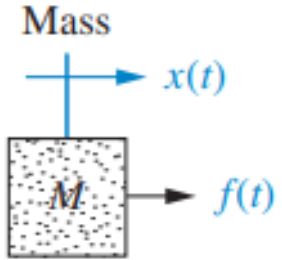
Contents

- Translational Mechanical Systems
- Rotational Mechanical Systems
- Transfer Functions for Systems with Gears

Translational Mechanical Systems

- There are analogies between electrical and mechanical components and variables.
- The transfer functions will be mathematically indistinguishable from an electrical network.
- Mechanical systems, like electrical networks, have three passive, linear components:
 - 1) viscous damper, dissipates energy, analogous to electrical resistor
 - 2) spring, energy-storage element, analogous to capacitor
 - 3) mass, energy-storage element, analogous to inductor

Translational Mechanical Systems

Component	Force-velocity	Force-displacement	Impedance $Z_M(s) = F(s)/X(s)$
 Spring	$f(t) = K \int_0^t v(\tau) d\tau$	$f(t) = Kx(t)$	K
 Viscous damper	$f(t) = f_v v(t)$	$f(t) = f_v \frac{dx(t)}{dt}$	$f_v s$
 Mass	$f(t) = M \frac{dv(t)}{dt}$	$f(t) = M \frac{d^2 x(t)}{dt^2}$	Ms^2

K , f_v , and M are called *spring constant*, *coefficient of viscous friction*, and *mass*, respectively

Translational Mechanical Systems

- Mechanical force is analogous to electrical voltage.
- Mechanical velocity is analogous to electrical current.
- Mechanical displacement is analogous to electrical charge.
- Thus, summing forces written in terms of velocity is analogous to summing voltages written in terms of current, and the resulting mechanical differential equations are analogous to **mesh** equations (vice versa gives nodal equations).
- If the forces are written in terms of displacement, the resulting mechanical equations resemble, but are not analogous to, the mesh equations. We, however, will use this model for mechanical systems so that we can write equations directly in terms of **displacement**.

Translational Mechanical Systems

The basic method. The basic method to find the transfer function of a mechanical system is writing related differential equation, called the *equation of motion*. To describe it:

Step 1: Assume a positive direction of motion, for example, to the right (This is similar to assuming a current direction in an electrical loop).

Step 2: Draw a free-body diagram, placing on the body all forces that act on the body either in the direction of motion or opposite to it.

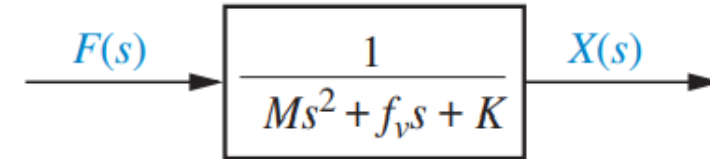
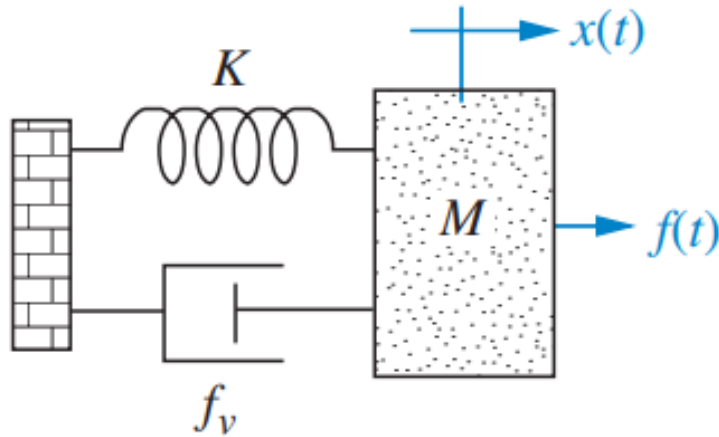
Step 3: Use Newton's law to form a differential equation of motion by summing the forces and setting the sum equal to zero.

Step 4: Take the Laplace transforms of the differential equations (assuming zero initial conditions).

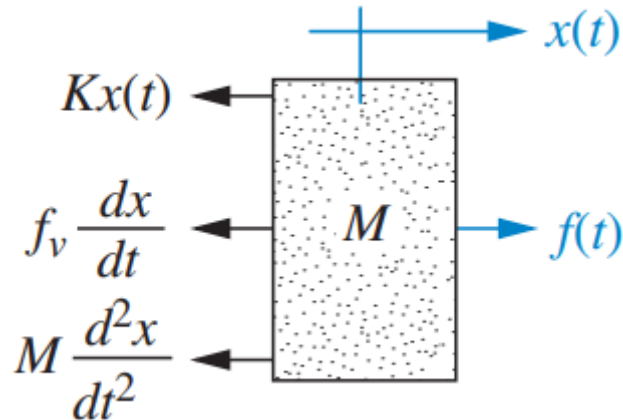
Step 5: Separate the variables, and arrive at the transfer function.

Translational Mechanical Systems

E.g. One Equation of Motion: Find the transfer function $X(s)/F(s)$.



Sln:



$$M \frac{d^2 x(t)}{dt^2} + f_v \frac{dx(t)}{dt} + Kx(t) = f(t)$$

$$Ms^2 X(s) + f_v s X(s) + KX(s) = F(s)$$

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{Ms^2 + f_v s + K}$$

Translational Mechanical Systems

The simplifying technique. Can we parallel our work with electrical networks by circumventing the writing of differential equations and by defining **impedances** for mechanical components?

- Taking the Laplace transform of the force-displacement equations, we obtain:

1) spring, $f(t) = Kx(t) \Rightarrow F(s) = KX(s)$

2) viscous damper, $f(t) = f_v \frac{dx(t)}{dt} \Rightarrow F(s) = f_v sX(s)$

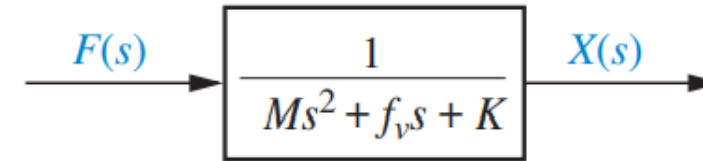
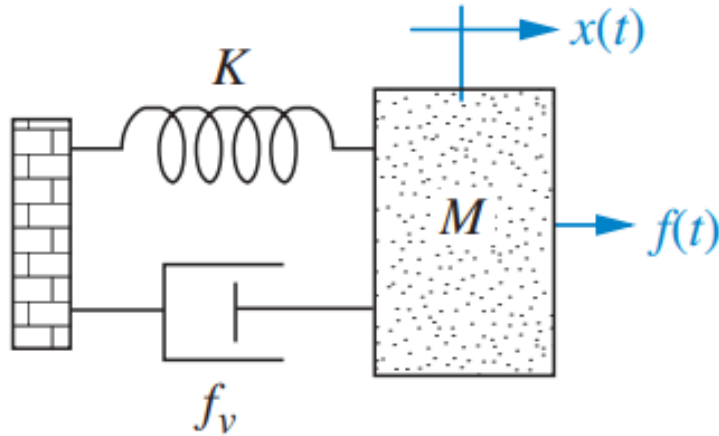
3) mass, $f(t) = M \frac{d^2x(t)}{dt^2} \Rightarrow F(s) = Ms^2X(s)$

and we define impedance for mechanical components as:

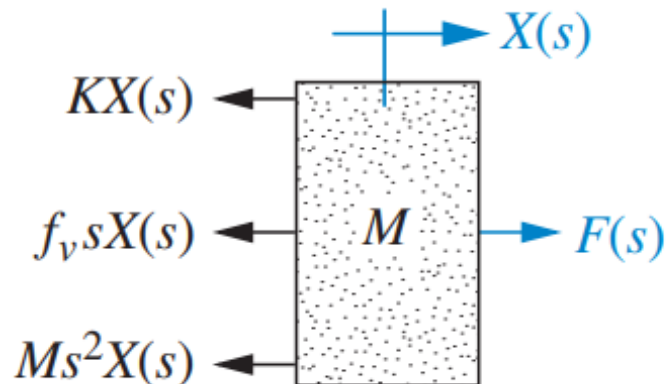
$$Z_M(s) = \frac{F(s)}{X(s)}$$

Translational Mechanical Systems

E.g. One Equation of Motion: Find the transfer function $X(s)/F(s)$.



Sln: Replacing each force by its Laplace transform, which is in the format $F(s) = Z_M(s)X(s)$.



$$Ms^2X(s) + f_v sX(s) + KX(s) = F(s)$$

$$(Ms^2 + f_v s + K)X(s) = F(s)$$

[Sum of impedances] $X(s)$ = [Sum of applied forces]
similar, but not analogous, to a mesh equation

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{Ms^2 + f_v s + K}$$

Translational Mechanical Systems

Multiple Degrees of Freedom.

- Many mechanical systems are similar to multiple-loop and -node electrical networks, where more than one simultaneous differential equation is required to describe the system.
- The number of motion equations required is equal to the **number of linearly independent motions**, that is called the **number of degrees of freedom**.
- **Linear independence** implies that a point of motion in a system can still move if all other points of motion are held still.

This is not meant to imply that these motions are not coupled to one another; in general, they are. For example, in a two-loop electrical network, each loop current depends on the other loop current, but if we open-circuit just one of the loops, the other current can still exist if there is a voltage source in that loop. Similarly, in a mechanical system with two degrees of freedom, one point of motion can be held still while the other point of motion moves under the influence of an applied force.

Translational Mechanical Systems

Multiple Degrees of Freedom.

In order to work such a problem, draw the free-body diagram for each point of motion and then use superposition.

Step 1: For each free-body diagram, hold all other points of motion still and find the forces acting on the body due only to its own motion.

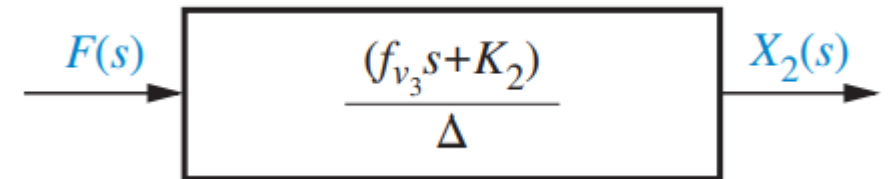
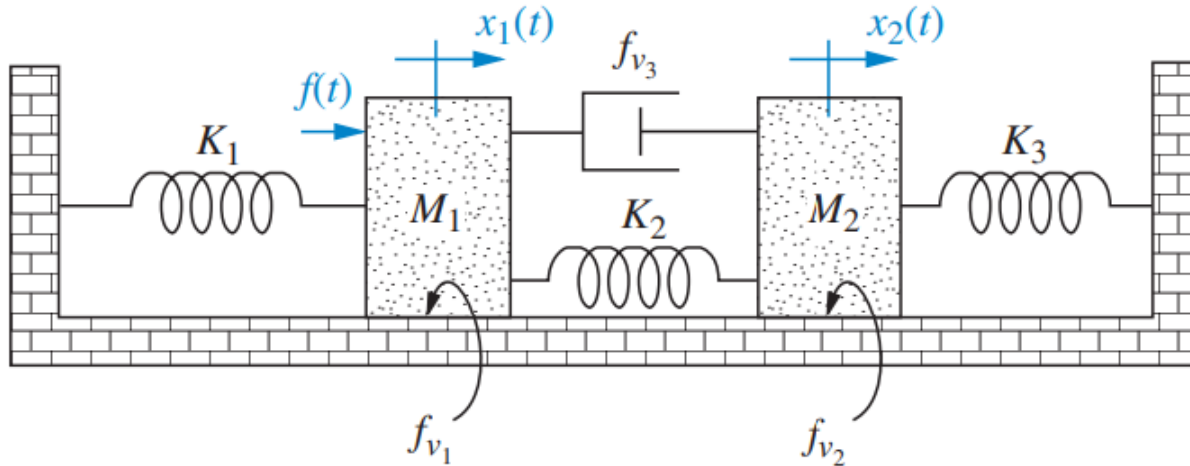
Step 2: Hold the body still and activate the other points of motion one at a time, placing on the original body the forces created by the adjacent motion.

Step 3: Using Newton's law, sum the forces on each body and set the sum to zero. The result is a system of simultaneous equations of motion.

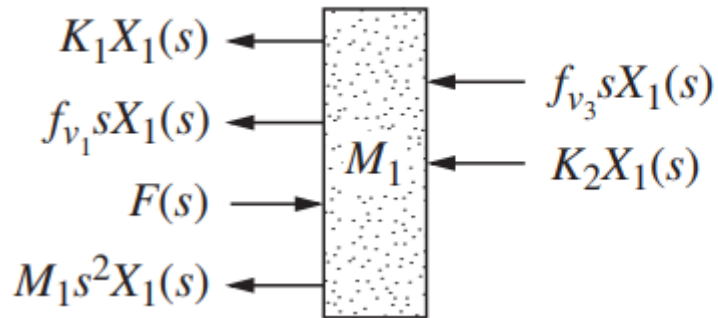
Step 4: Solve for the transfer function.

Translational Mechanical Systems

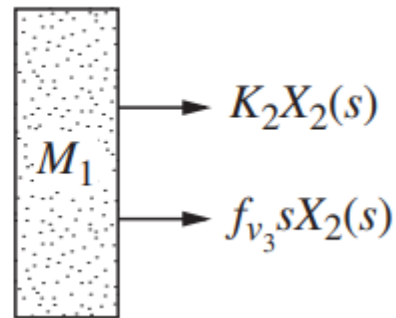
E.g. Two Degrees of Freedom: Find the transfer function $X_2(s)/F(s)$.



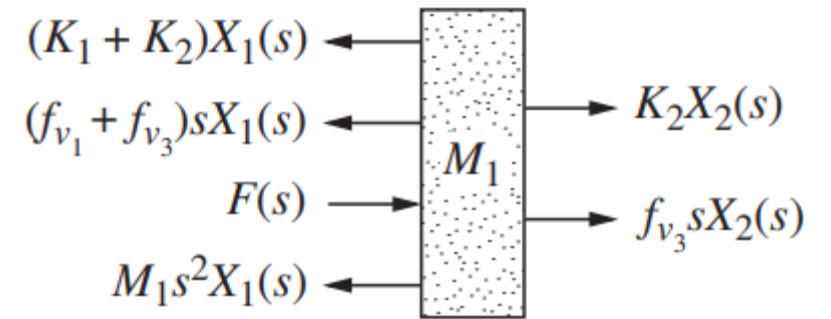
Sln:



a. Forces on M_1 due only to motion of M_1



b. forces on M_1 due only to motion of M_2

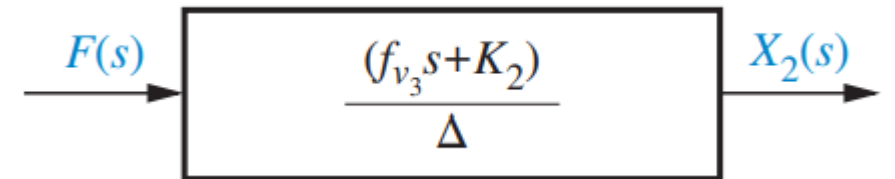
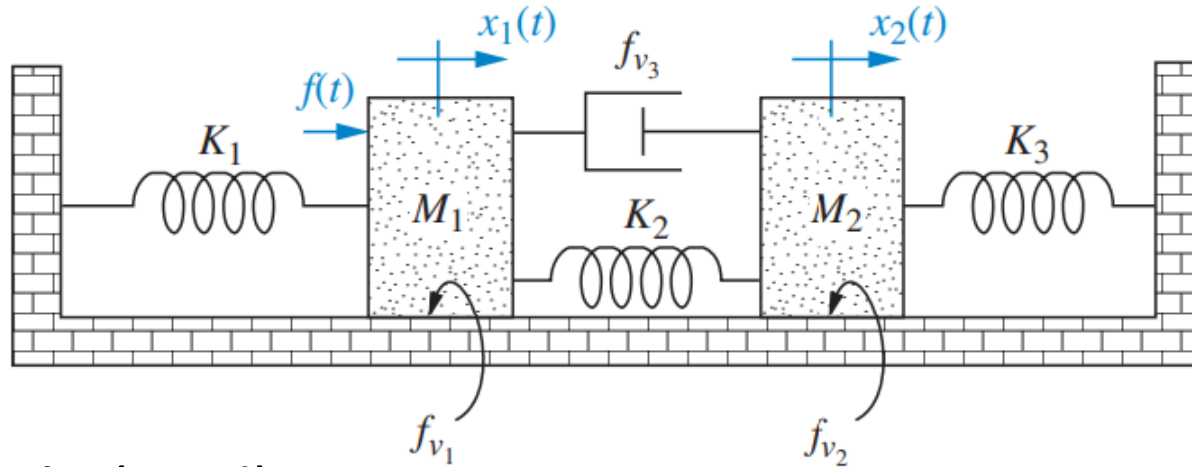


c. all forces on M_1

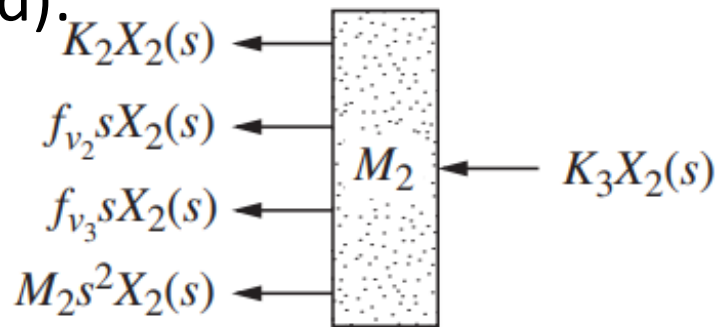
$$[M_1 s^2 (f_{v1} + f_{v3})s + (K_1 + K_2)]X_1(s) - (f_{v3}s + K_2)X_2(s) = F(s)$$

Translational Mechanical Systems

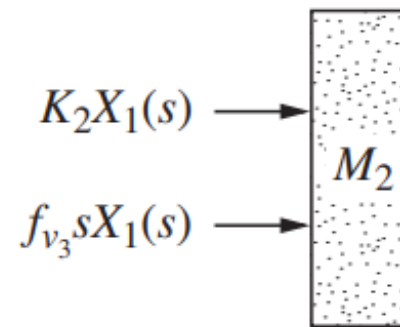
E.g. Two Degrees of Freedom: Find the transfer function $X_2(s)/F(s)$.



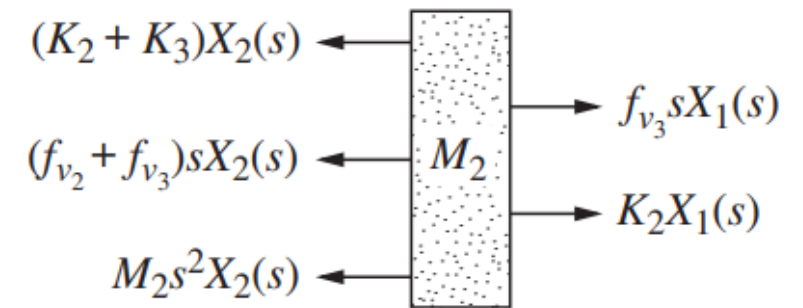
Sln (cntd):



a. Forces on M_2 due only to motion of M_2



b. forces on M_2 due only to motion of M_1

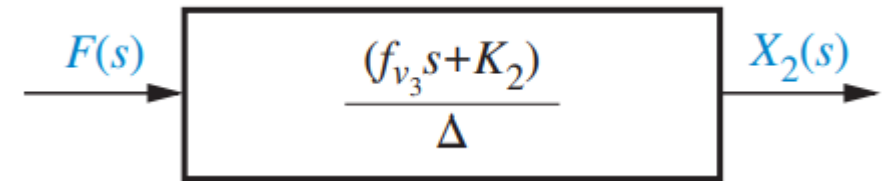
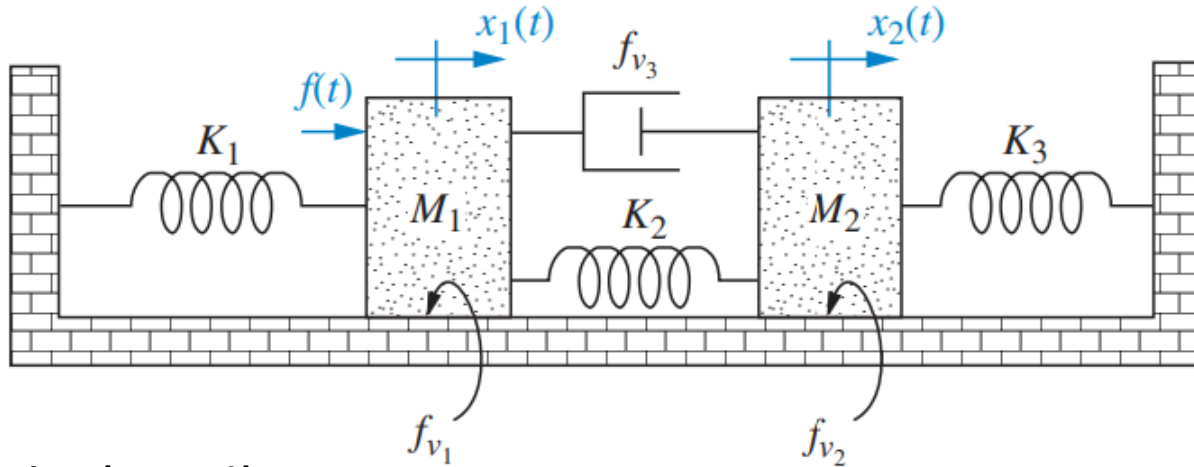


c. all forces on M_2

$$-(f_{v3}s + K_2)X_1(s) + [M_2s^2 + (f_{v2} + f_{v3})s + (K_2 + K_3)]X_2(s) = 0$$

Translational Mechanical Systems

E.g. Two Degrees of Freedom: Find the transfer function $X_2(s)/F(s)$.



Sln (cntd):

The Laplace transform of the motion equations:

$$[M_1 s^2 (f_{v1} + f_{v3})s + (K_1 + K_2)]X_1(s) - (f_{v3}s + K_2)X_2(s) = F(s)$$

$$-(f_{v3}s + K_2)X_1(s) + [M_2 s^2 + (f_{v2} + f_{v3})s + (K_2 + K_3)]X_2(s) = 0$$

The transfer function:

$$\frac{X_2(s)}{F(s)} = G(s) = \frac{(f_{v3}s + K_2)}{\Delta}$$

$$\Delta = \begin{vmatrix} [M_1 s^2 + (f_{v1} + f_{v3})s + (K_1 + K_2)] & -(f_{v3}s + K_2) \\ -(f_{v3}s + K_2) & [M_2 s^2 + (f_{v2} + f_{v3})s + (K_2 + K_3)] \end{vmatrix}$$

Translational Mechanical Systems

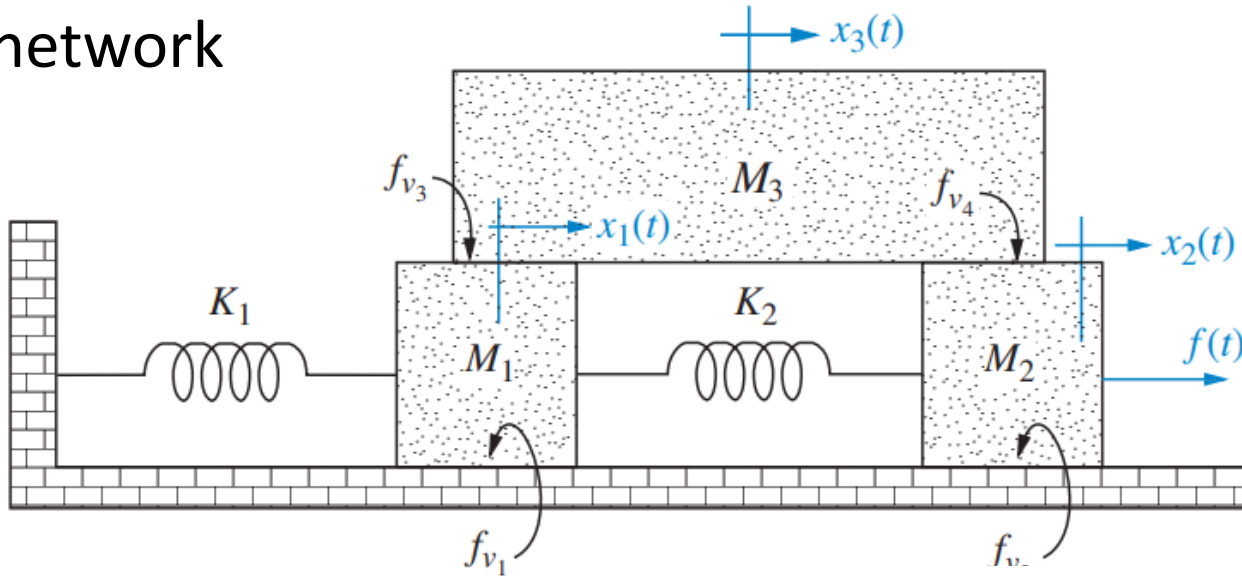
Multiple Degrees of Freedom.

An alternative method, without drawing the free-body diagram, similar to electrical mesh equations:

$$\begin{aligned}
 & \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{connected} \\ \text{to the motion} \\ \text{at } x_1 \end{array} \right] X_1(s) - \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{between} \\ x_1 \text{ and } x_2 \end{array} \right] X_2(s) - \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{between} \\ x_1 \text{ and } x_3 \end{array} \right] X_3(s) = \left[\begin{array}{c} \text{Sum of} \\ \text{applied forces} \\ \text{at } x_1 \end{array} \right] \\
 & - \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{between} \\ x_1 \text{ and } x_2 \end{array} \right] X_1(s) + \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{connected} \\ \text{to the motion} \\ \text{at } x_2 \end{array} \right] X_2(s) - \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{between} \\ x_2 \text{ and } x_3 \end{array} \right] X_3(s) = \left[\begin{array}{c} \text{Sum of} \\ \text{applied forces} \\ \text{at } x_2 \end{array} \right] \\
 & - \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{between} \\ x_1 \text{ and } x_3 \end{array} \right] X_1(s) - \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{between} \\ x_2 \text{ and } x_3 \end{array} \right] X_2(s) + \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{connected} \\ \text{to the motion} \\ \text{at } x_3 \end{array} \right] X_3(s) = \left[\begin{array}{c} \text{Sum of} \\ \text{applied forces} \\ \text{at } x_3 \end{array} \right]
 \end{aligned}$$

Translational Mechanical Systems

E.g. Three Degrees of Freedom - Equations of Motion by Inspection: Write, but do not solve, the equations of motion for the following mechanical network



$$[M_1 s^2 + (f_{v1} + f_{v3})s + (K_1 + K_2)]X_1(s) - K_2 X_2(s) - f_{v3} s X_3(s) = 0$$

$$-K_2 X_1(s) + [M_2 s^2 + (f_{v2} + f_{v4})s + K_2]X_2(s) - f_{v4} s X_3(s) = F(s)$$

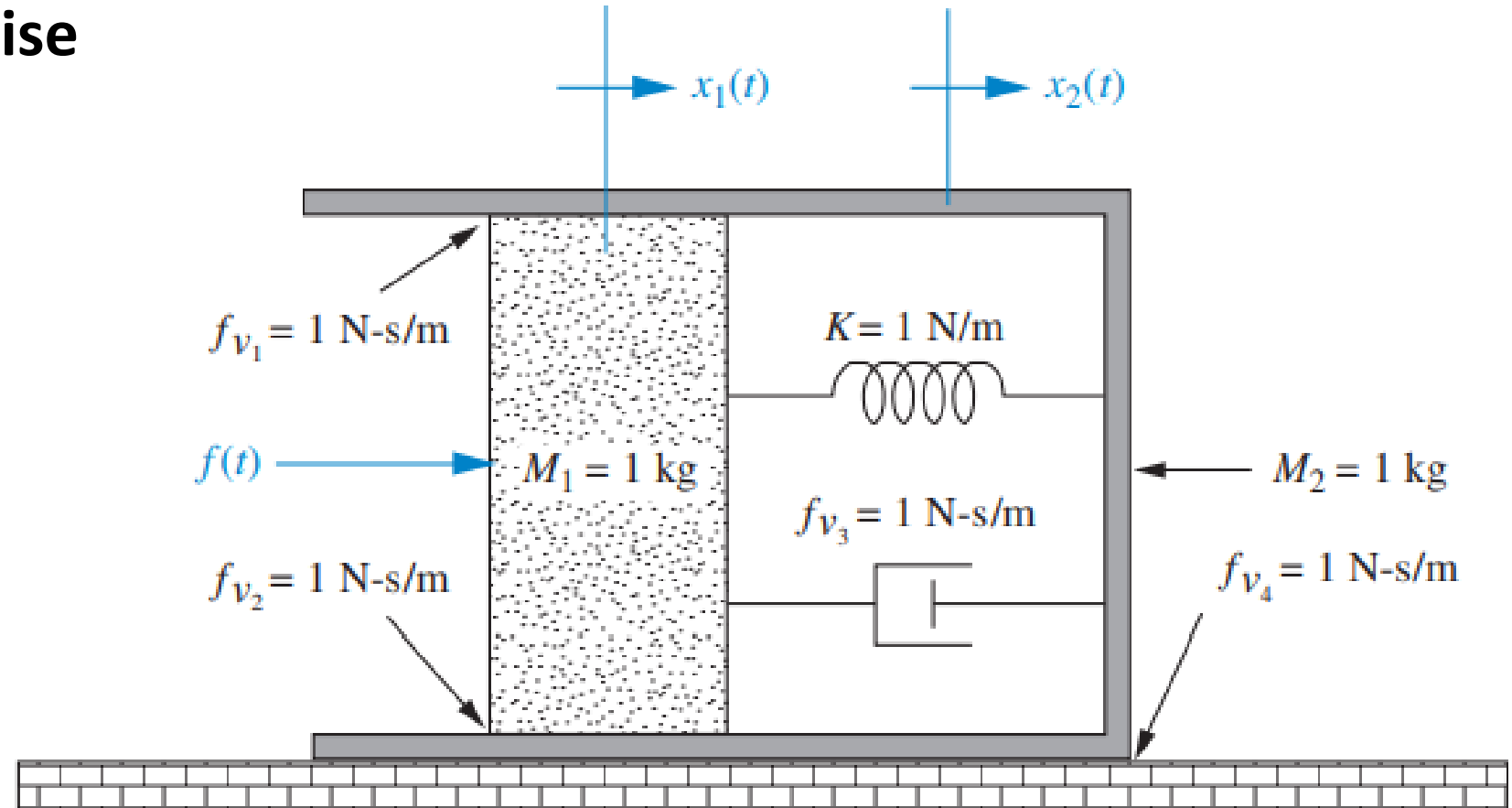
$$-f_{v3} s X_1(s) - f_{v4} s X_2(s) + [M_3 s^2 + (f_{v3} + f_{v4})s]X_3(s) = 0$$

Translational Mechanical Systems

- **Skill-Assessment Exercise**

Find the transfer function

$$G(s) = X_2(s)/F(s).$$

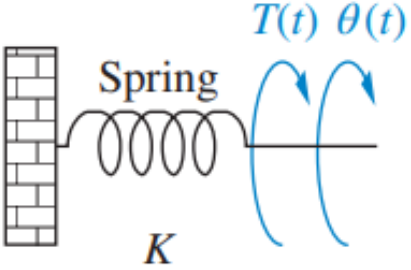
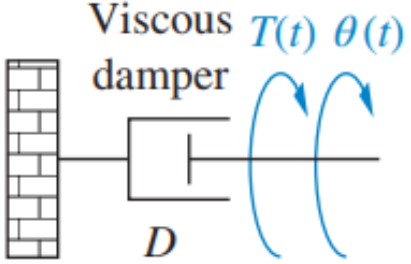
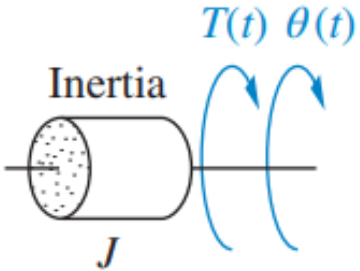


ANSWER:
$$G(s) = \frac{3s + 1}{s(s^3 + 7s^2 + 5s + 1)}$$

Rotational Mechanical Systems

- Rotational mechanical systems are handled the same way as translational mechanical systems, except that
 - the components undergo rotation instead of translation,
 - torque replaces force,
 - angular displacement replaces translational displacement,
 - angular velocity replaces translational velocity
 - mass is replaced by **inertia**.

Rotational Mechanical Systems

Component	Torque-angular velocity	Torque-angular displacement	Impedance $Z_M(s) = T(s)/\theta(s)$
	$T(t) = K \int_0^t \omega(\tau) d\tau$	$T(t) = K\theta(t)$	K
	$T(t) = D\omega(t)$	$T(t) = D \frac{d\theta(t)}{dt}$	Ds
	$T(t) = J \frac{d\omega(t)}{dt}$	$T(t) = J \frac{d^2\theta(t)}{dt^2}$	Js^2

The values of K , D , and J are called *spring constant*, *coefficient of viscous friction*, and *moment of inertia*, respectively.

Rotational Mechanical Systems

Degrees of Freedom.

The concept of degrees of freedom carries over to rotational systems, except that

- We test a point of motion by rotating it while holding still all other points of motion.
- The number of points of motion that can be rotated while all others are held still equals the number of equations of motion required to describe the system.

Rotational Mechanical Systems

Degrees of Freedom.

Writing the equations of motion for rotational systems is similar to writing them for translational systems; the only difference is that the free-body diagram consists of torques rather than forces. We obtain these torques using superposition. For each free-body diagram,

Step 1: rotate a body while holding all other points still and place on its free-body diagram all torques due to the body's own motion..

Step 2: hold the body still, we rotate adjacent points of motion one at a time and add the torques due to the adjacent motion to the free-body diagram.

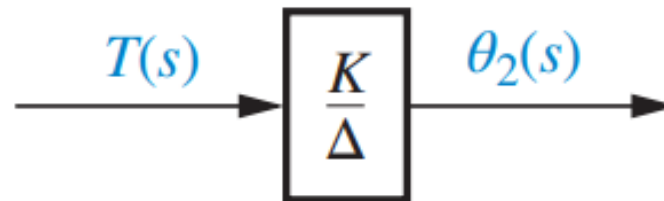
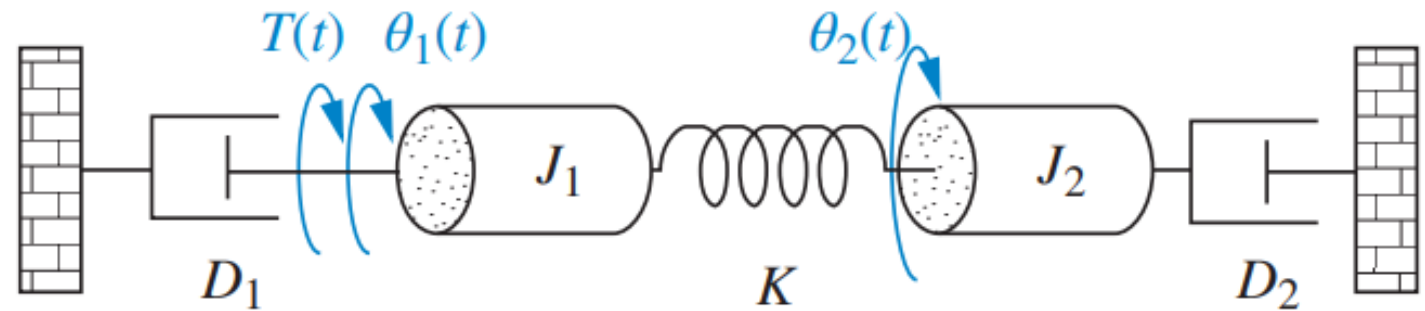
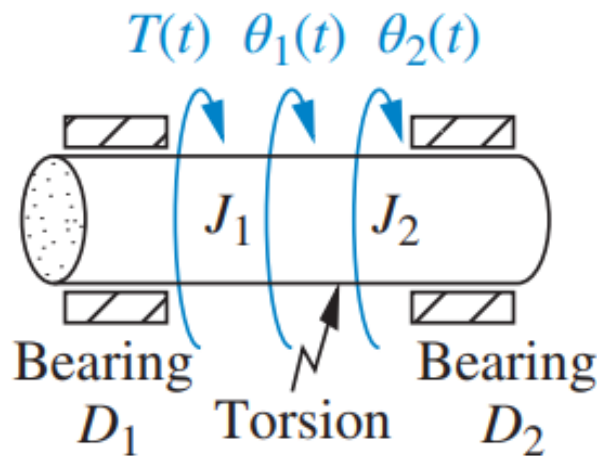
Step 3: sum these torques and set that equal to zero to form the motion equation.

Step 4: repeat the process for each point of motion.

Step 5: Solve for the transfer function.

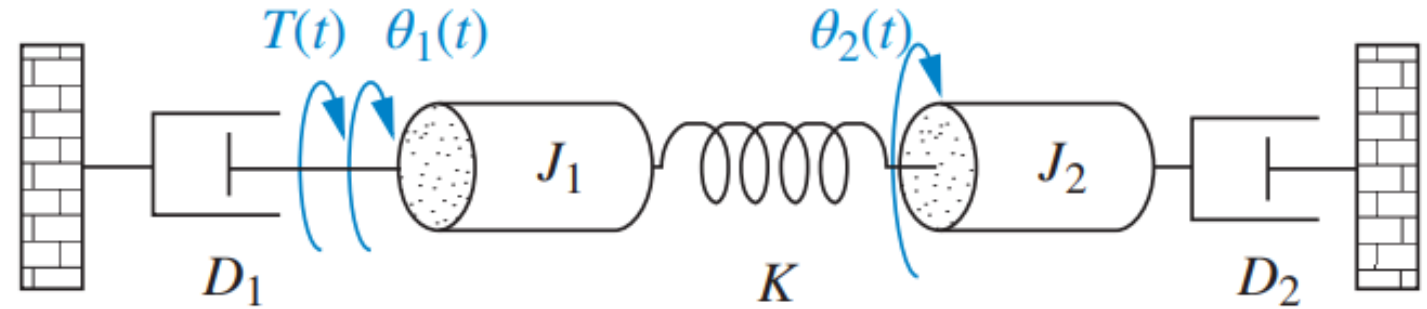
Rotational Mechanical Systems

E.g. Two Degrees of Freedom – Free-Body Diagrams: Find the transfer function $\theta_2(s)/T(s)$. The rod is supported by bearings at either end and is undergoing torsion. A torque is applied at the left, and the displacement is measured at the right.



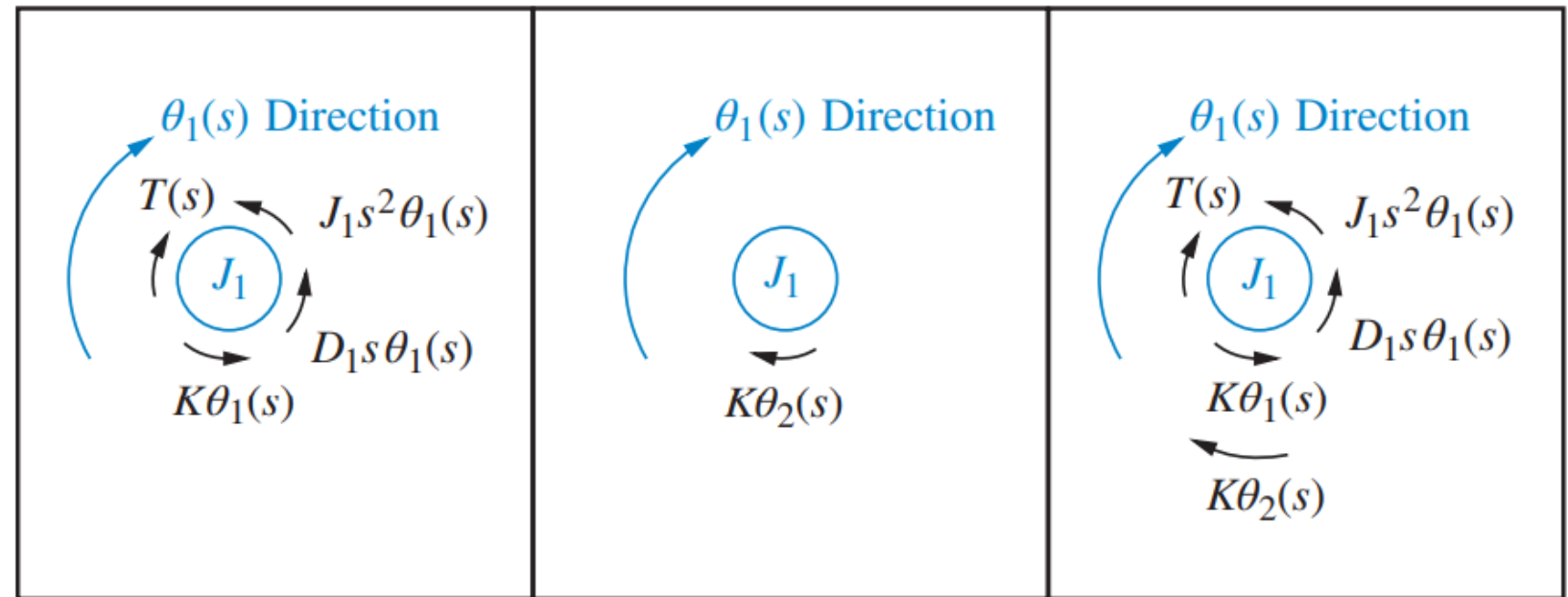
Rotational Mechanical Systems

E.g. (Cntd) Two Degrees of Freedom – Free-Body Diagrams:
Find the transfer function $\theta_2(s)/T(s)$.



Sln:

Free-body diagram for J_1

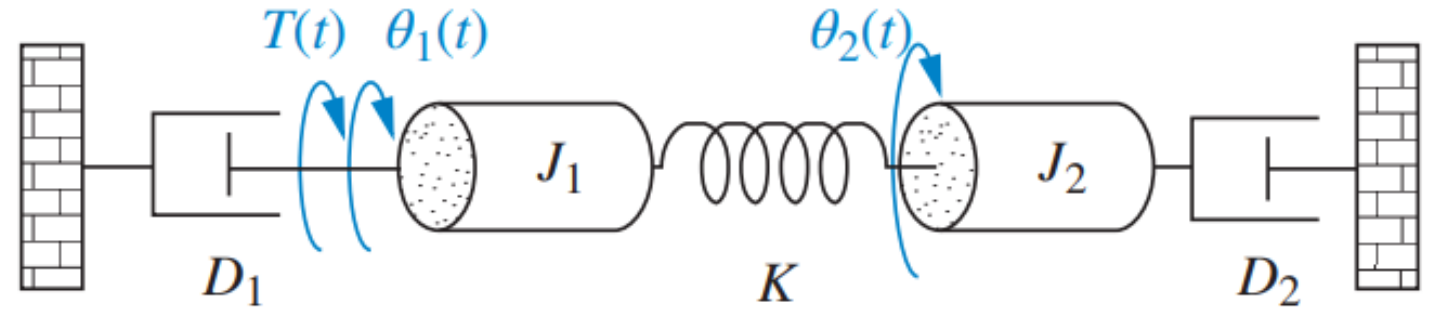


$$(J_1 s^2 + D_1 s + K) \theta_1(s)$$

$$- K \theta_2(s) = T(s)$$

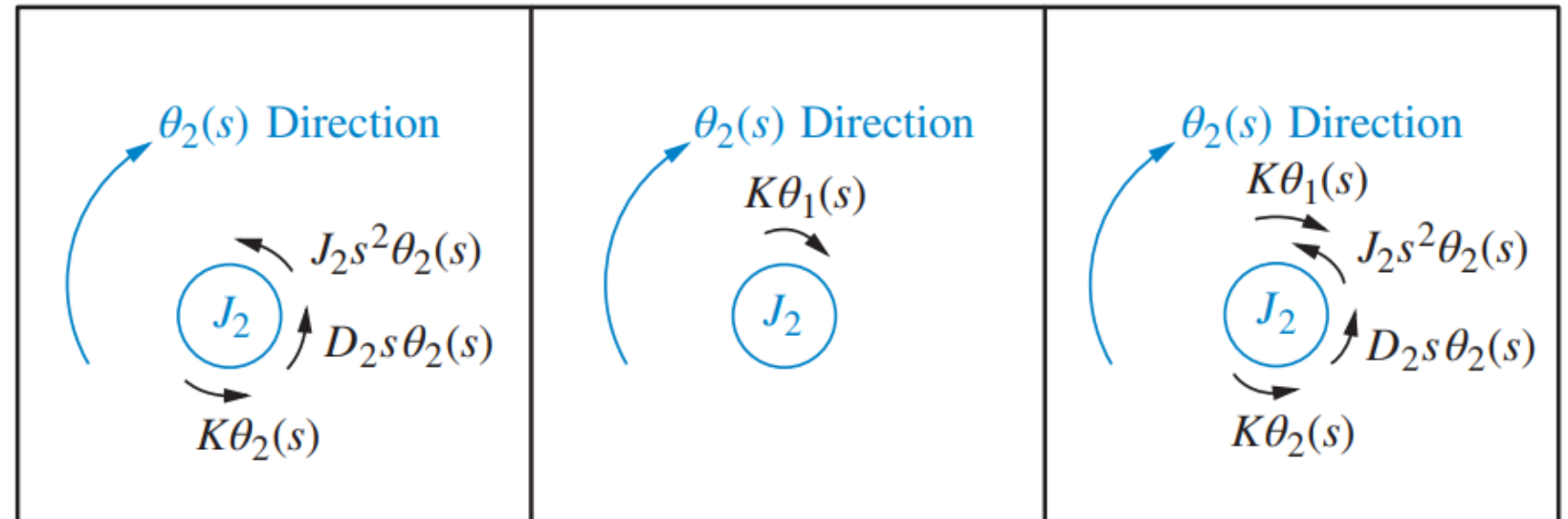
Rotational Mechanical Systems

E.g. (Cntd) Two Degrees of Freedom – Free-Body Diagrams:
Find the transfer function $\theta_2(s)/T(s)$.



Sln (Cntd):

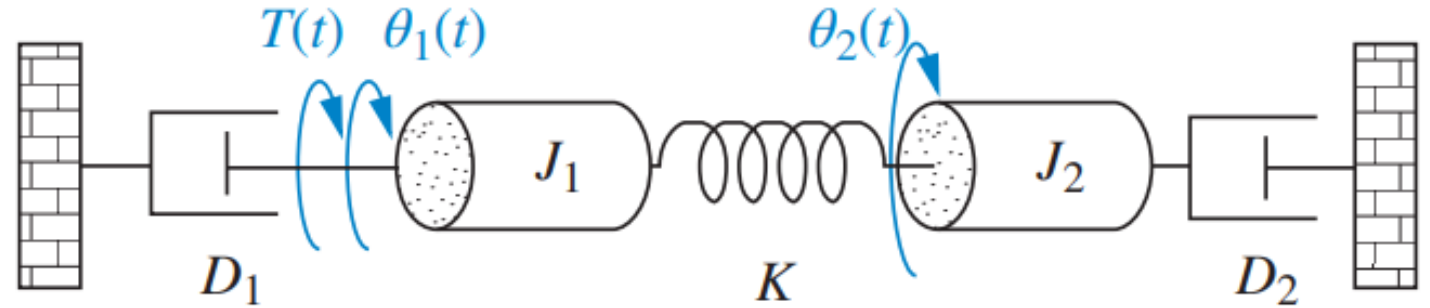
Free-body diagram for J_2



$$-K\theta_1(s) + (J_2s^2 + D_2s + K)\theta_2(s) = 0$$

Rotational Mechanical Systems

E.g. (Cntd) Two Degrees of Freedom – Free-Body Diagrams:
Find the transfer function $\theta_2(s)/T(s)$.



Sln (Cntd):

The motion equations:

$$(J_1 s^2 + D_1 s + K)\theta_1(s) - K\theta_2(s) = T(s)$$

$$-K\theta_1(s) + (J_2 s^2 + D_2 s + K)\theta_2(s) = 0$$

The transfer function:

$$\frac{\theta_2(s)}{T(s)} = \frac{K}{\Delta}$$

$$\Delta = \begin{vmatrix} (J_1 s^2 + D_1 s + K) & -K \\ -K & (J_2 s^2 + D_2 s + K) \end{vmatrix}$$

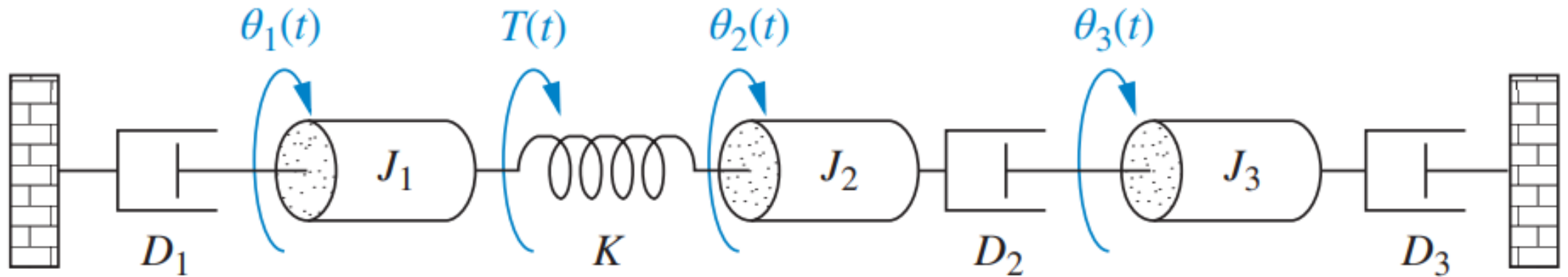
Rotational Mechanical Systems

Multiple Degrees of Freedom – by Inspection:

$$\begin{aligned} & \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{connected} \\ \text{to the motion} \\ \text{at } \theta_1 \end{array} \right] \theta_1(s) - \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{between} \\ \theta_1 \text{ and } \theta_2 \end{array} \right] \theta_2(s) = \left[\begin{array}{c} \text{Sum of} \\ \text{applied torques} \\ \text{at } \theta_1 \end{array} \right] \\ & - \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{between} \\ \theta_1 \text{ and } \theta_2 \end{array} \right] \theta_1(s) + \left[\begin{array}{c} \text{Sum of} \\ \text{impedances} \\ \text{connected} \\ \text{to the motion} \\ \text{at } \theta_2 \end{array} \right] \theta_2(s) = \left[\begin{array}{c} \text{Sum of} \\ \text{applied torques} \\ \text{at } \theta_2 \end{array} \right] \end{aligned}$$

Rotational Mechanical Systems

E.g. Three Degrees of Freedom – by Inspection: Write, but do not solve, the Laplace transform of the equations of motion.



$$(J_1 s^2 + D_1 s + K)\theta_1(s) \quad -K\theta_2(s) \quad -0\theta_3(s) = T(s)$$

$$-K\theta_1(s) + (J_2 s^2 + D_2 s + K)\theta_2(s) \quad -D_2 s\theta_3(s) = 0$$

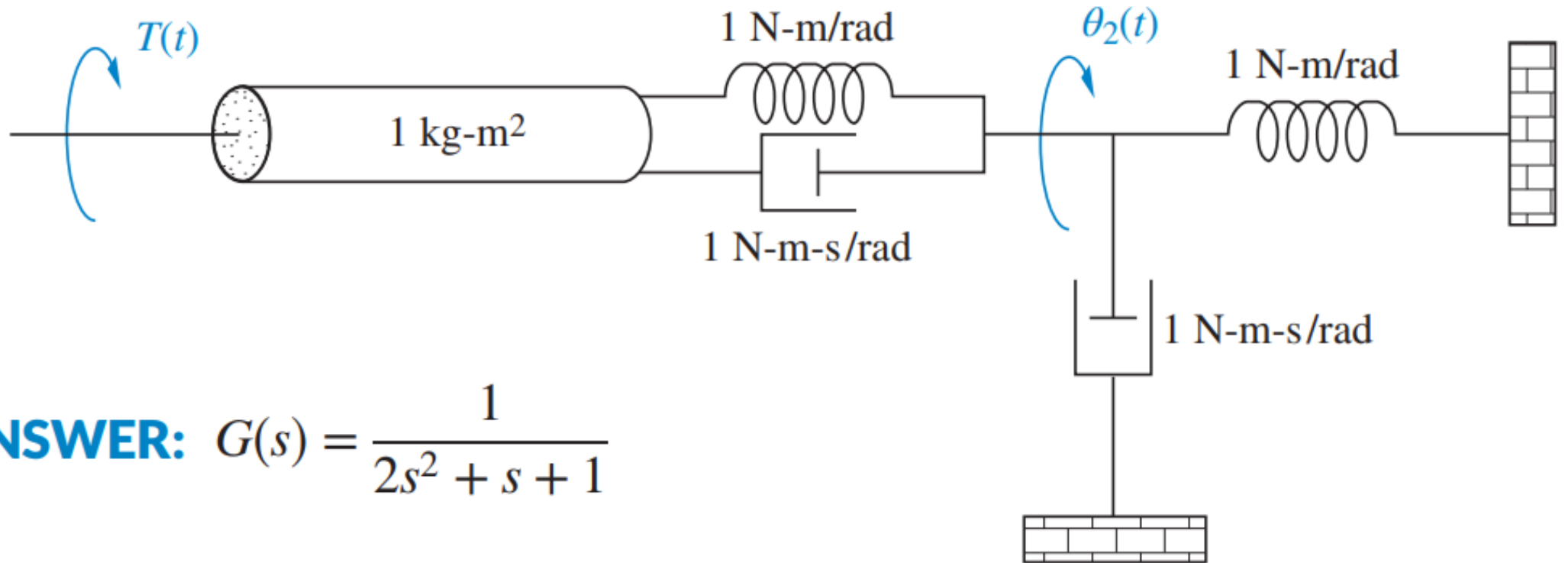
$$-0\theta_1(s) \quad -D_2 s\theta_2(s) + (J_3 s^2 + D_3 s + D_2 s)\theta_3(s) = 0$$

Rotational Mechanical Systems

- **Skill-Assessment Exercise**

Find the transfer function

$$G(s) = \theta_2(s)/T(s).$$



ANSWER: $G(s) = \frac{1}{2s^2 + s + 1}$

Transfer Functions for Systems with Gears

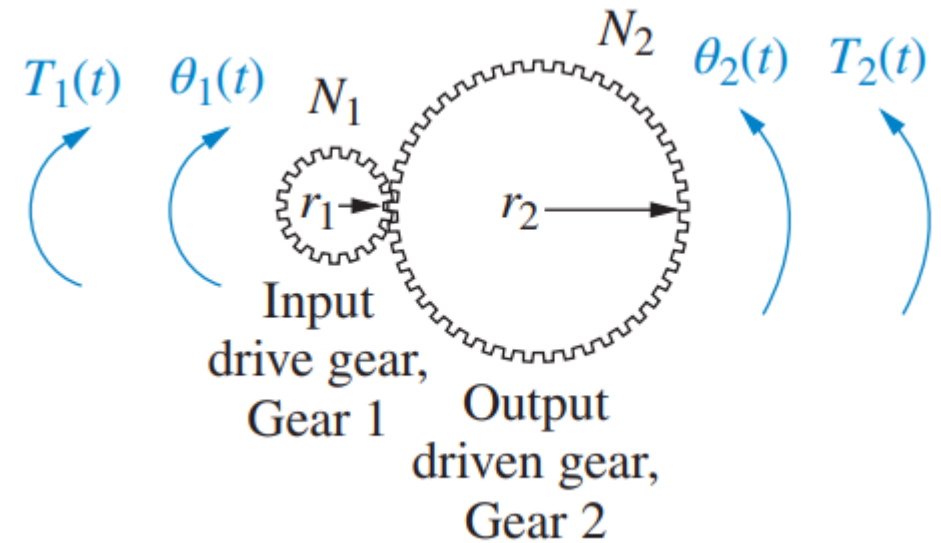
- Rotational systems, especially those driven by motors, are rarely seen without associated gear trains driving the load. Let's cover this important topic.
- Gears provide mechanical advantage to rotational systems. Consider the effect of gearing in riding bicycle.
 - Going uphill, you shift to provide more torque and less speed.
 - On the straightaway, you shift to obtain more speed and less torque.

Thus, gears allow you to match the drive system and the load—a trade-off between speed and torque.

- In this section, we idealize the behavior of gears and assume that there is **no backlash**.

Transfer Functions for Systems with Gears

- The linearized interaction between two gears is considered here.
- An input gear with radius r_1 and N_1 teeth is rotated through angle θ_1 due to a torque, T_1 .
- An output gear with radius r_2 and N_2 teeth responds by rotating through angle θ_2 and delivering a torque, T_2 .



Transfer Functions for Systems with Gears

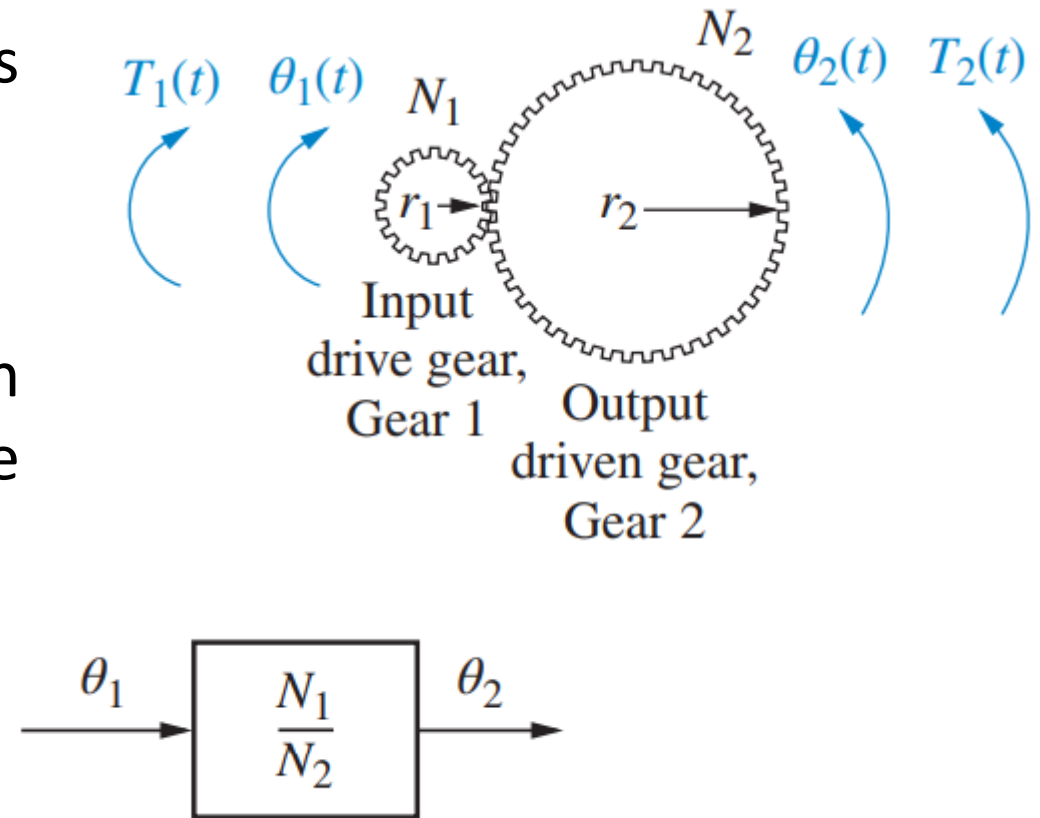
The relationship between the rotation, θ , of two interacted gears:

- The distance traveled along each gear's circumference is the same.

$$r_1\theta_1 = r_2\theta_2$$

Since the ratio of the number of teeth along the circumference is in the same proportion as the ratio of the radii.

$$\frac{\theta_2}{\theta_1} = \frac{r_1}{r_2} = \frac{N_1}{N_2}$$

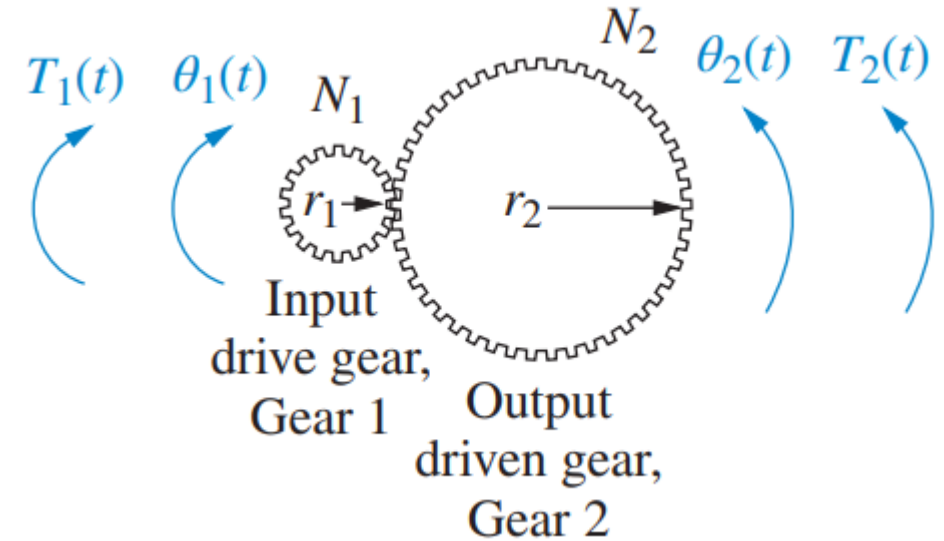


The ratio of the angular displacement of the gears is **inversely** proportional to the ratio of the number of teeth.

Transfer Functions for Systems with Gears

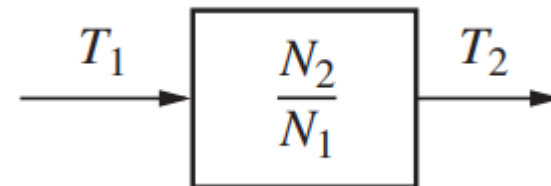
The relationship between the input torque, T_1 , and the delivered torque, T_2 :

- Assume the gears are lossless, that is, they do not absorb or store energy, the energy into Gear 1 equals the energy out of Gear 2.
- Since the translational energy of force times displacement becomes the rotational energy of torque times angular displacement.



$$T_1 \theta_1 = T_2 \theta_2$$

$$\frac{T_2}{T_1} = \frac{\theta_1}{\theta_2} = \frac{N_2}{N_1}$$



The torques are **directly proportional** to the ratio of the number of teeth.

Transfer Functions for Systems with Gears

Reflecting the mechanical impedances from the output to the input, and eliminating the gears

- T_1 can be reflected to the output

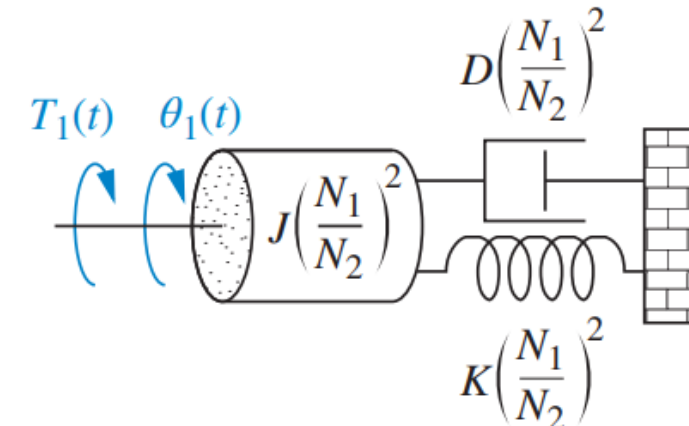
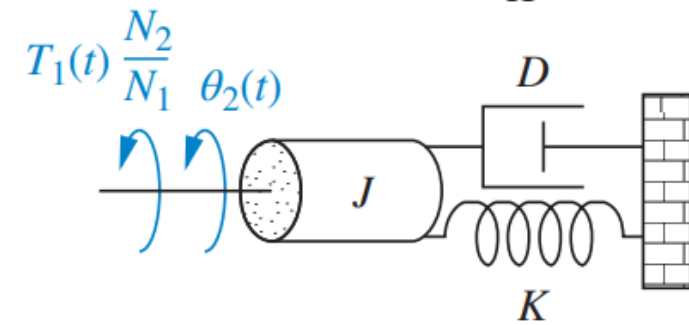
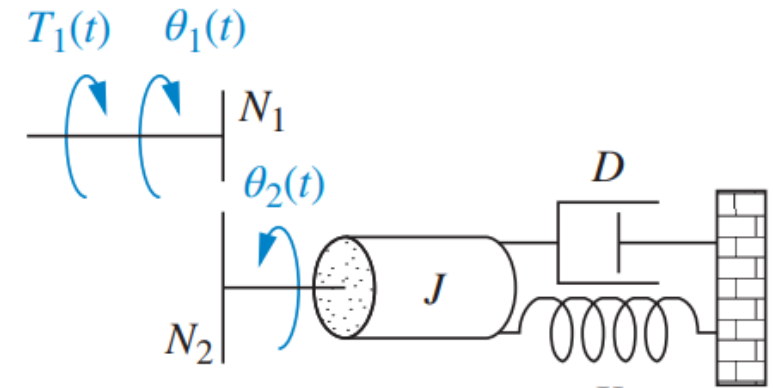
$$(Js^2 + Ds + K) \theta_2(s) = T_1(s) \frac{N_2}{N_1}$$

- Convert θ_2 into an equivalent θ_1

$$(Js^2 + Ds + K) \frac{N_1}{N_2} \theta_1(s) = T_1(s) \frac{N_2}{N_1}$$

- After simplification

$$\left[J \left(\frac{N_1}{N_2} \right)^2 s^2 + D \left(\frac{N_1}{N_2} \right)^2 s + K \left(\frac{N_1}{N_2} \right)^2 \right] \theta_1(s) = T_1(s)$$



Transfer Functions for Systems with Gears

Reflecting the mechanical impedances from the output to the input, and eliminating the gears

Generalizing the results:

Rotational mechanical impedances can be reflected through gear trains by multiplying by the ratio:

$$\left(\frac{\text{Number of teeth of gear on } \textit{destination} \text{ shaft}}{\text{Number of teeth of gear on } \textit{source} \text{ shaft}} \right)^2$$

where the impedance to be reflected is attached to the **source** shaft and is being reflected to the **destination** shaft.

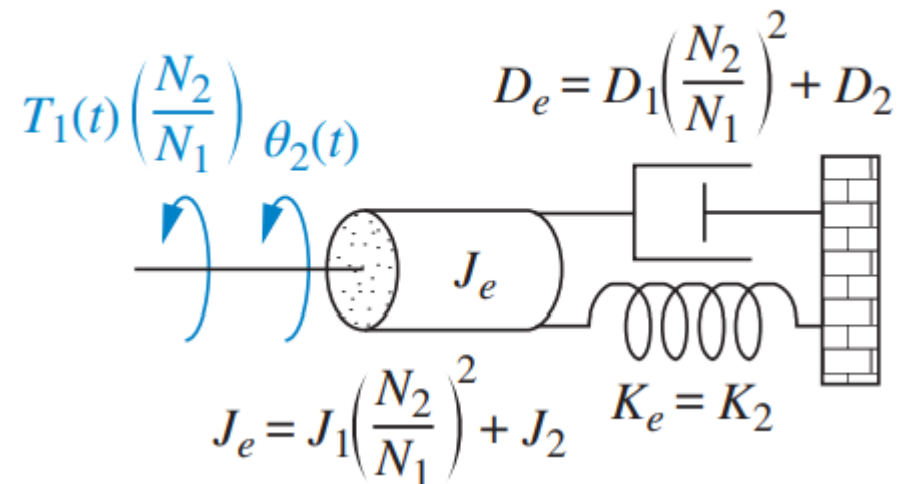
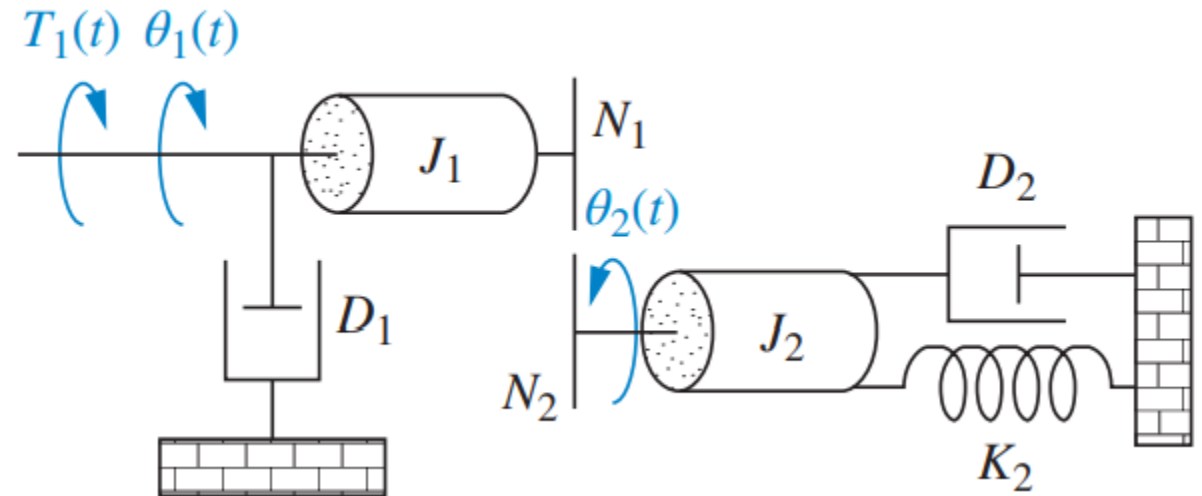
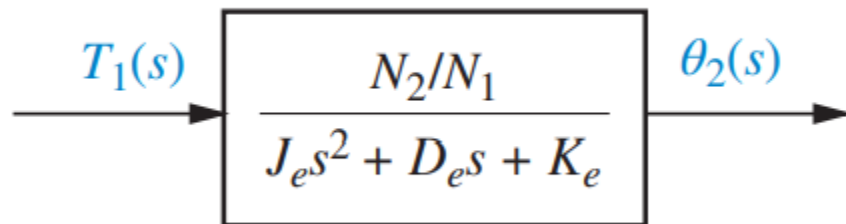
Transfer Functions for Systems with Gears

E.g. System with Lossless Gears: Find the transfer function $\theta_2(s)/T_1(s)$.

Sln:

The inertias, do not undergo linearly independent motion, since they are tied together by the gears. Thus, there is only one degree of freedom and hence one equation of motion.

$$(J_e s^2 + D_e s + K_e) \theta_2(s) = T_1(s) \frac{N_2}{N_1}$$

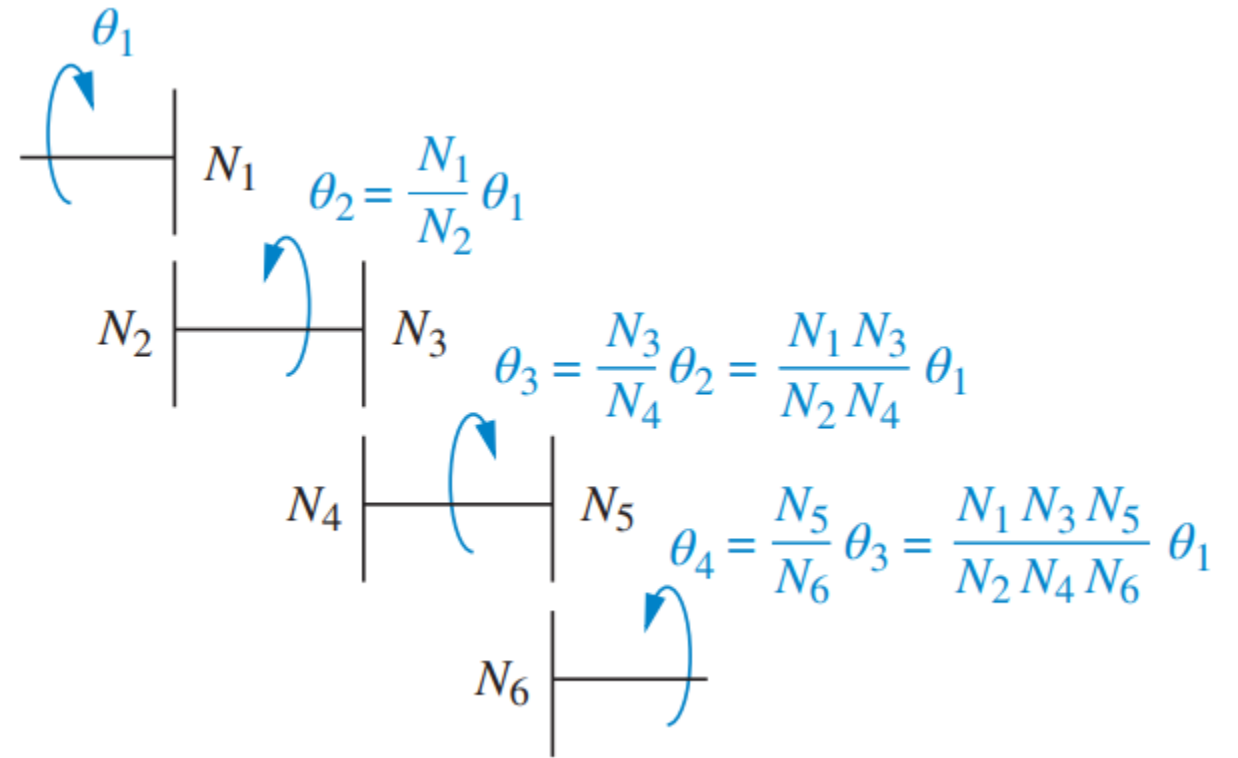


Transfer Functions for Systems with Gears

Gear train:

In order to eliminate gears with large radii, a gear train is used to implement large gear ratios by cascading smaller gear ratios.

$$\theta_4 = \frac{N_1 N_3 N_5}{N_2 N_4 N_6} \theta_1$$

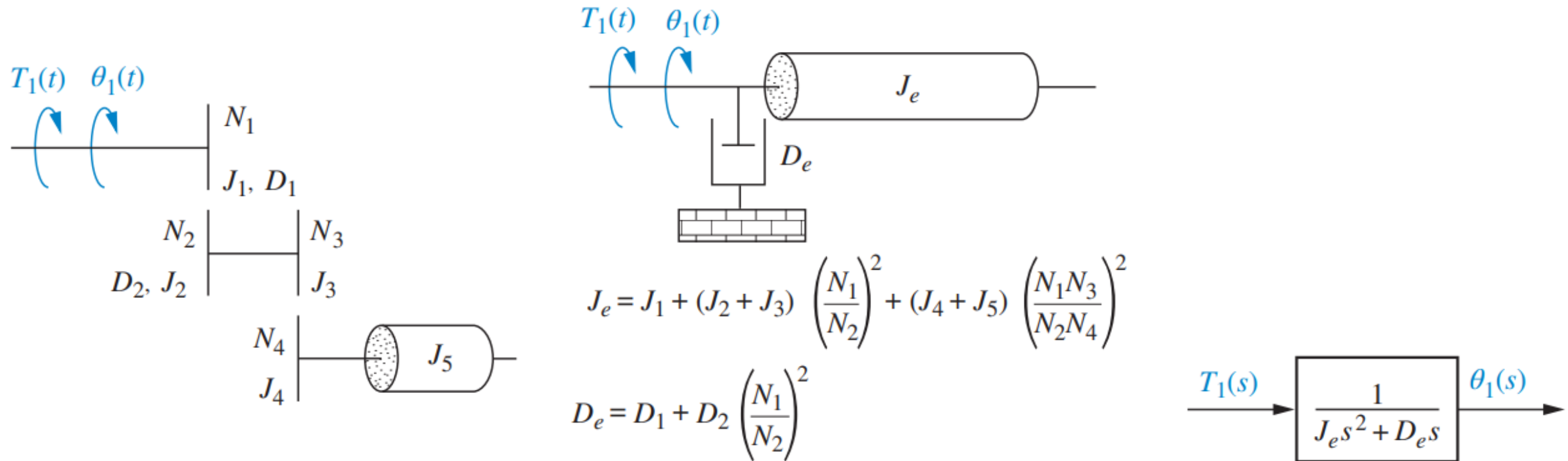


The equivalent gear ratio is the product of the individual gear ratios.

Transfer Functions for Systems with Gears

Transfer Function—Gears with Loss

PROBLEM: Find the transfer function, $\theta_1(s)/T_1(s)$,



Appendix

Cramer's Rule

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

$$\text{Let } D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

If $D \neq 0$ then

$$x = \frac{\begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}}{D}$$

$$y = \frac{\begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}}{D}$$

$$z = \frac{\begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}}{D}$$